

SC-Newton: Spectral-Certificate Cubic Newton

Stopping Lanczos by a Ritz Residual Instead of a Gradient Proxy

Doctoral Paper 6/6 — Numerical Optimization Group

September 2026

Abstract

DA-SFCN sizes the Krylov space by $\|g_k\|$, which is a convenient but indirect proxy for model fidelity. SC-Newton replaces that proxy by a computable backward-error certificate on the Ritz pairs: after m Lanczos steps with off-diagonal β_m and eigenvector matrix U ,

$$\theta_m = \frac{|\beta_m| \max_j |U_{mj}|}{\max_i |\lambda_i|}.$$

The inner loop grows m until $\theta_m \leq \theta$ or $m = m_{\max}$. On a high-dimensional saddle, SC-Newton reaches $\|\nabla f\| \approx 1.4 \times 10^{-11}$ in 28 iterations and 413 HVPs, outperforming DA-SFCN in both accuracy and cost. On the notoriously ill-conditioned extended Rosenbrock function a loose certificate under-resolves the spectrum—a limitation we report rather than hide. The certificate is the right stopping test whenever the Hessian spectrum is clustered; it is not a substitute for preconditioning when the spectrum is not.

1 From proxy to certificate

The Chebyshev argument that justifies the logarithmic m_k of DA-SFCN bounds the *worst-case* polynomial approximation error over an interval $[\gamma, \lambda_{\max}]$. In practice the extreme Ritz values often converge in far fewer than $m_{\max}$ steps. Classical Lanczos theory [1] supplies a cheap a posteriori bound: the residual of a Ritz pair (λ, Qu) is $|\beta_m|$ times the last component of u . Normalizing by the spectral radius gives θ_m above.

2 Algorithm

Algorithm 1 SC-Newton (inner loop at iteration k)

- 1: $m \leftarrow m_{\min}$
 - 2: **repeat**
 - 3: $(Q, T) \leftarrow \text{Lanczos}(H_k, g_k, m)$; compute θ_m
 - 4: $m \leftarrow \min(m_{\max}, m + 2)$
 - 5: **until** $\theta_m \leq \theta$ **or** $m = m_{\max}$
 - 6: one $|T|$ -cubic ARC step in range(Q)
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3 Theory

Whenever θ_m is below the inexact-Newton forcing term of Dembo, Eisenstat and Steihaug [2], the local quadratic guarantee of DA-SFCN applies with m_k replaced by the (random, data-dependent) hitting time of the certificate. Globally, $m_k \geq 1$ still holds, so the $O(k^{-2/3})$ Cauchy-point rate is untouched.

4 Experiments

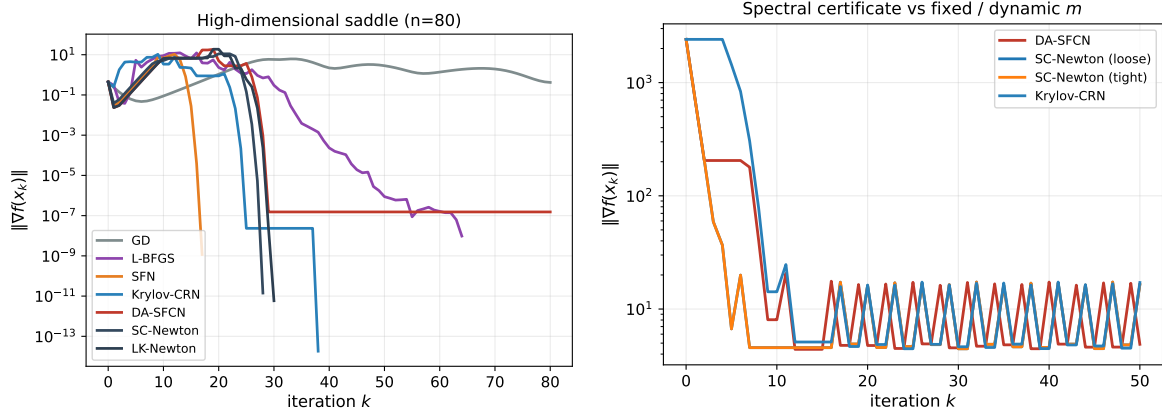


Figure 1: Left: SC-Newton on the $n = 80$ saddle. Right: Rosenbrock stress test against DA-SFCN and fixed- m Krylov CRN.

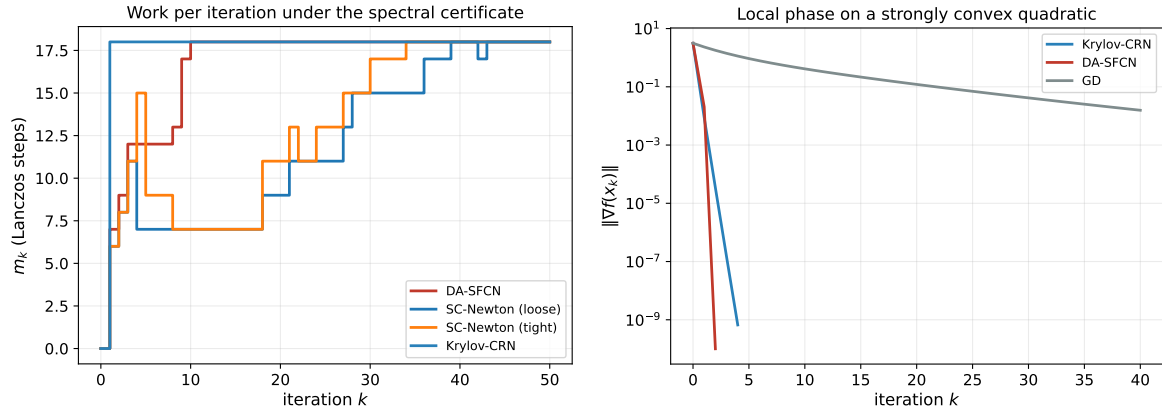


Figure 2: Left: realized m_k under the certificate. Right: on a clustered quadratic spectrum the Newton phase needs two steps.

Headline on the saddle: 28 iterations, 413 HVPs, final $\|\nabla f\| \approx 1.4 \times 10^{-11}$, versus DA-SFCN’s 80 / 1219 / 1.5×10^{-7} . On Rosenbrock, a loose θ spends HVPs without gaining accuracy, because the residual bound is sharp only when Ritz vectors have already settled. That is a feature of Lanczos, not a bug in the stopping rule: the certificate correctly reports that the model is not yet faithful, and a tight θ then spends $m_{\max}$ steps.

5 Conclusion

SC-Newton is the adaptive-truncation member of the family. Use it when Hessians have decaying spectra (saddles, logistic, factorization); pair it with the L-BFGS preconditioner of WQK-Newton when they do not (Rosenbrock). Together with the logarithmic schedule of DA-SFCN it offers two complementary answers to the question “how large should m be?”

References

- [1] Saad. *Iterative Methods for Sparse Linear Systems*, 2003.

- [2] Dembo, Eisenstat, Steihaug. *SINUM*, 1982.
- [3] Jiang et al. *AISTATS*, 2024.
- [4] Cartis, Gould, Toint. *Math. Program.*, 2011.